

Writing Equations from Word Situations

Writing and solving one-step equations from a story, writing and solving two-step equations, and writing an expression from a story and evaluating it for a given value

How to turn a situation into an equation.

- Say in one sentence what the letter stands for, including its unit — “let n be the number of months”. An unnamed letter is where most errors start.
- Find the amount that happens once (a fee, a starting amount) and the amount that happens per unit (a rate). A rate is always multiplied by the variable; a one-time amount is added or subtracted.
- Write the equation before you compute anything, and make sure both sides measure the same kind of thing.
- Solve by undoing the operations in reverse order, then check the solution in the original sentence, not in your rearranged version.

1. Maya bought 6 packs of pencils. Each pack holds the same number of pencils, and she ended up with 72 pencils in all. Write an equation for the number of pencils in one pack, then solve it.

Answer: _____ pencils

2. A club’s fund held some money, in dollars. The club spent 45 on supplies, and 130 was left. Write an equation for the amount the fund held before the purchase, then solve it.

Answer: _____ dollars

3. A taxi charges a flat 3, in dollars, plus 2 for each mile driven. Write an expression for the fare of a ride of m miles, then use it to find the fare for a ride of 7 miles.

Answer: _____ dollars

4. A gym charges a one-time joining fee of 20, in dollars, plus 15 each month. Ren has paid 110 in all. Write an equation for the number of months Ren has been a member, then solve it.

Answer: _____ months

5. A rectangular garden bed has a perimeter of 46 cm in a scale drawing, and its width is 9 cm. Write an equation for the length of the bed, then solve it.

Answer: _____ cm

6. A muffin recipe uses $\frac{3}{4}$ cup of oats for one batch. A baker used 6 cups of oats in all, with none left over. Write an equation for the number of batches, then solve it.

Answer: _____ batches

7. A pool holds 300 gallons of water and is draining at 12 gallons per minute. Write an expression for the number of gallons still in the pool after t minutes, then use it to find how much water is left after 18 minutes.

Answer: _____ gallons

8. A restaurant takes 6, in dollars, off a bill with a coupon. Three friends then split what is left evenly, and each of them pays 14. Write an equation for the bill before the coupon, then solve it.

Answer: _____ dollars

9. A rink rents skates for 5, in dollars, plus 3 for each hour. Ana paid 23 in all. Ben modelled this with the equation $5h + 3 = 23$, where h is the number of hours. Explain in one sentence why Ben's equation does not match the situation, then write the correct equation and solve it.

Answer: _____ hours

10. Two thirds of the members of a coding club signed up for a contest. Then 4 of those who signed up dropped out, leaving 18 contestants. Write an equation for the number of members in the club, then solve it.

Answer: _____ students